Temporal and Departmental Determinants of Service Closure Times: A Statistical Analysis of Boston 311 Service Requests

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Introduction

Contemporary municipal governance increasingly relies upon 311 non-emergency service platforms to facilitate citizen-government communication and coordinate urban service delivery. Boston's 311 infrastructure processes substantial daily volumes of constituent requests, generating rich datasets suitable for analytical investigation aimed at operational enhancement and citizen satisfaction improvement (Hastie et al., 2009).

This study analyzes five years (2015-2019) of Boston's 311 service data to address three fundamental research questions: First, which temporal dynamics characterize service fulfillment patterns across seasonal cycles and annual progressions? Second, how do organizational unit characteristics and service request taxonomies influence resolution duration and schedule adherence? Third, can statistical learning algorithms successfully classify service requests vulnerable to processing delays, thereby enabling anticipatory management intervention?

The dataset comprises approximately 1.2 million service transactions, averaging 657 daily submissions (see Table 1). Preliminary assessment indicates an aggregate closure rate of 93.05% coupled with schedule compliance rate of 79.47%, suggesting substantial opportunity space for Service Level Agreement (SLA) performance enhancement. Understanding the fundamental drivers underlying these performance indicators enables precision-targeted interventions and optimal resource allocation strategies (City of Boston, 2020).

Table 1. Boston 311 Dataset Summary (2015-2019)

Metric	Value
Time Period	2015-2019
Total Service Requests	1,199,625
Average Daily Volume	657
Closure Rate (%)	93.05%
On-Time Performance (%)	79.47%

Methods

Our analytical framework synthesizes multiple complementary statistical techniques to generate comprehensive operational intelligence. The methodology was selected to provide both descriptive insights and predictive capabilities, thereby supporting tactical operational decisions alongside strategic organizational planning.

Data Preparation and Transformation

Data preprocessing involved several critical transformations. First, temporal variables were created including season classification (Spring: March-May, Summer: June-August, Fall: September-November, Winter: December-February), weekend indicators, and yearly categorical variables. Second, a binary outcome variable was constructed for logistic regression, classifying cases as "fast closure" (≤5 days) versus "slow closure" (>5 days). Third, continuous closure time in days was calculated for linear regression modeling. These transformations enable comprehensive examination of temporal patterns and facilitate predictive model development.

Analytical Methods

- Exploratory Subset Analysis: Descriptive statistics were computed across multiple categorical dimensions including temporal factors (year, season), organizational characteristics (department), and service taxonomies (request type). This analysis reveals patterns in case volume, closure rates, on-time performance, and median resolution times across different operational contexts.
- Binary Logistic Regression: Logistic regression models were developed to classify service requests as fast (≤5 days) versus slow (>5 days) resolution. Four nested models were estimated: (1) temporal factors only (weekend, season, year), (2) adding department effects, (3) incorporating request type, and (4) the full model including neighborhood and submission source. Model performance was evaluated using accuracy, precision, recall, and F1 scores on a holdout test set. This hierarchical approach allows assessment of incremental predictive value added by each variable category.
- Linear Regression: Ordinary least squares (OLS) regression was applied to predict continuous closure time in days. A parallel nested modeling strategy was employed,

systematically adding predictor categories to evaluate their contribution to explained variance (R^2). This approach quantifies the magnitude and direction of effects for continuous outcomes.

- Penalized Regression (LASSO): Least Absolute Shrinkage and Selection Operator (LASSO) regression was implemented for automated variable selection and model simplification. LASSO applies L1 regularization to shrink less important coefficients toward zero, effectively performing feature selection. Nine candidate predictors were evaluated including temporal metrics (hour, day, week), operational volume measures (daily and hourly case counts), and aggregated performance indicators (neighborhood, department, and request type average closure times). Cross-validation was used to identify the optimal penalty parameter ($\lambda = 0.024$), balancing model complexity against predictive accuracy.
- Justification of Methods: These methods were selected for their complementary strengths. Exploratory subset analysis provides intuitive, actionable insights for operational stakeholders without statistical expertise. Logistic regression addresses the practical classification problem of identifying at-risk cases for intervention. Linear regression quantifies continuous effects and provides interpretable coefficients. LASSO regression automates feature selection and prevents overfitting when numerous predictors exist. Together, this integrated approach addresses both the descriptive question (what patterns exist?) and the predictive question (can we forecast delays?), while maintaining interpretability essential for operational decision-making (James et al., 2013).

Results

Temporal Patterns: Annual Trends

Analysis of annual patterns reveals notable temporal dynamics in Boston's 311 service delivery. Table 2 presents summary statistics disaggregated by year, showing case volumes ranging from 210,081 (2015) to 262,742 (2018). Closure rates remained consistently high across all years (91.0%-95.0%), demonstrating stable operational capacity. However, on-time performance exhibited marked improvement from 69.71% in 2015 to 83.01% in 2016, followed by slight decline in subsequent years (80.07% by 2019). Median closure times decreased substantially from 1.00 days in 2015 to 0.32-0.39 days in 2016-2019, indicating accelerated processing following initial year operations.

Figure 1. Case Performance Trends by Year (2015-2019)

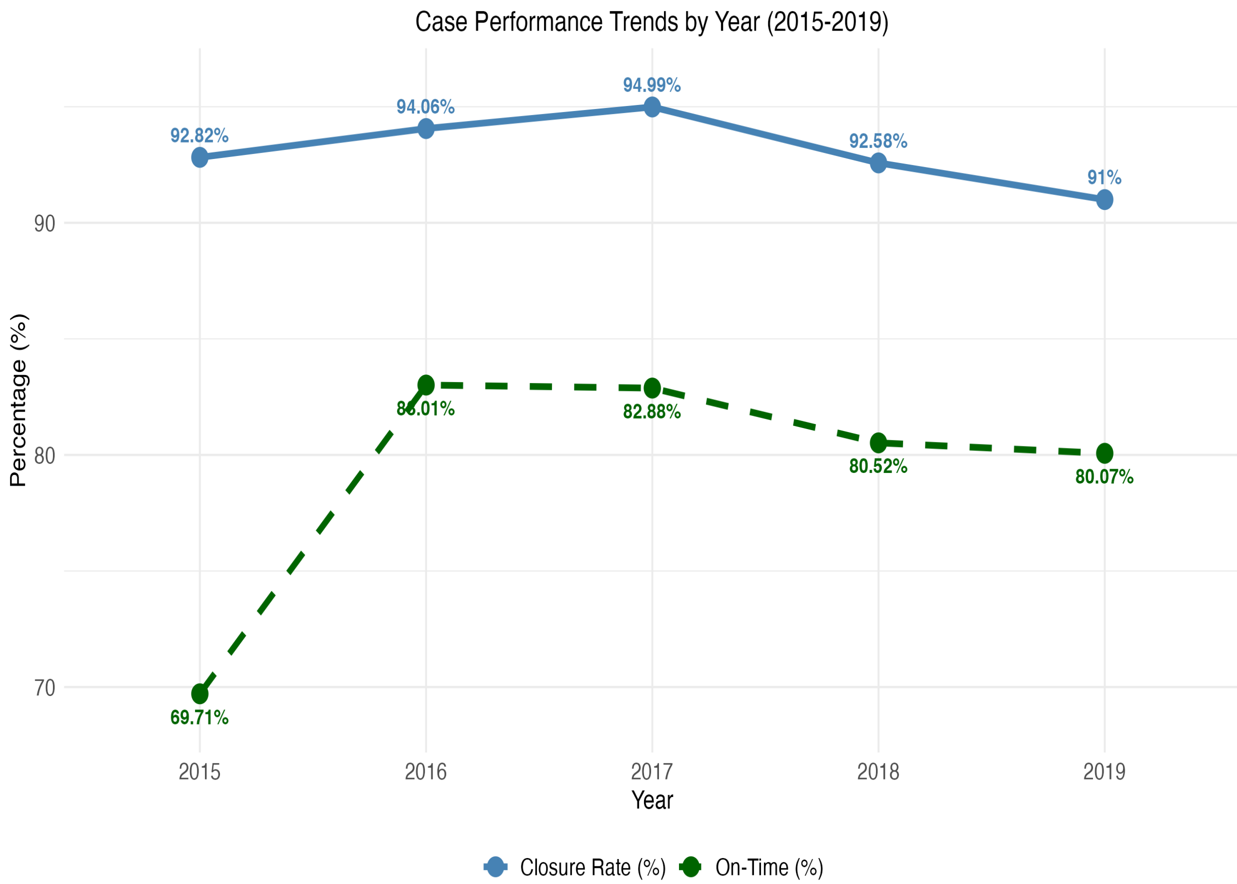


Table 2. Service Request Summary by Year

Year	Number of Cases	Closure Rate	On-Time (%)	Median Closure Days
2015	210081	92.82%	69.71%	1
2016	216526	94.06%	83.01%	0.32
2017	251258	94.99%	82.88%	0.39
2018	262742	92.58%	80.52%	0.39
2019	259018	91%	80.07%	0.36

Temporal Patterns: Seasonal Variation

Seasonal analysis (Table 3) demonstrates moderate variation in service request volume and performance metrics. Summer months generated the highest case volume (319,491), while winter months showed slightly elevated closure rates (93.46%). On-time performance varied from 75.02% in winter to 81.81% in fall, suggesting weather-related or resource allocation effects. Median closure times ranged from 0.38 days (fall) to 0.58 days (winter), with winter months consistently exhibiting longer resolution periods.

Figure 2. Median Closure Days by Season

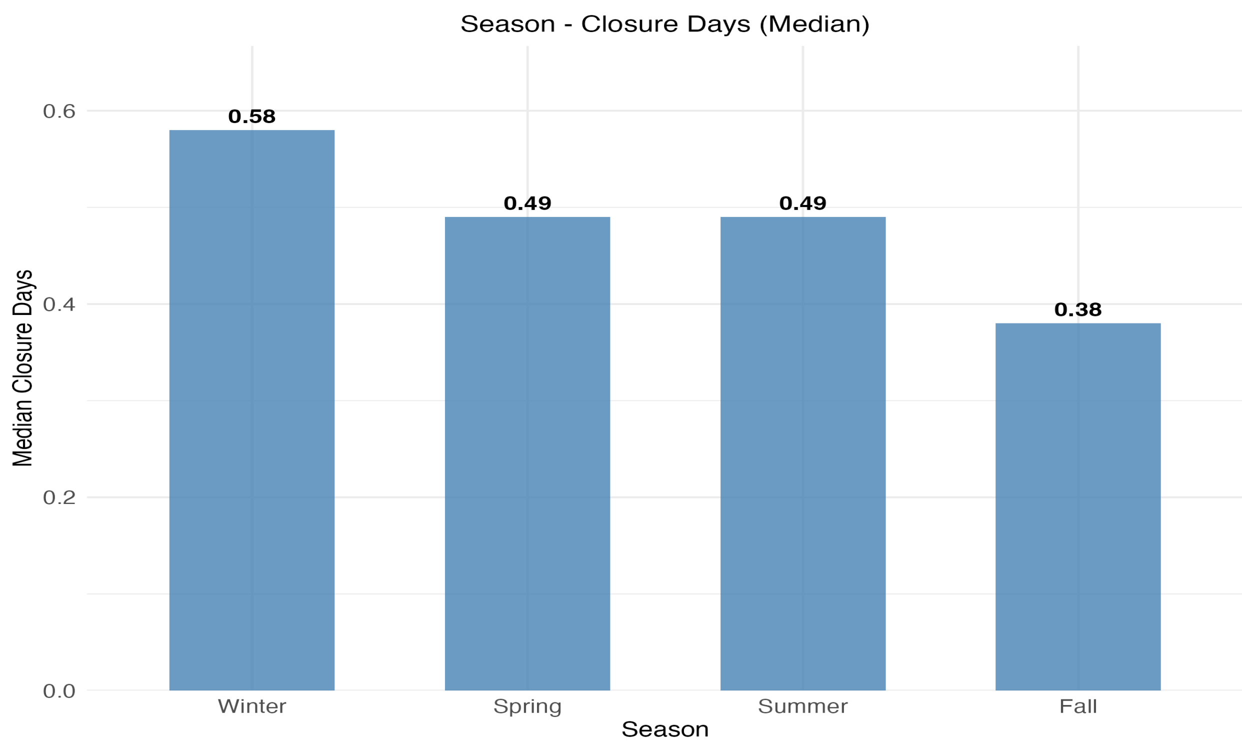


Table 3. Service Request Summary by Season

Season	Cases	Closure Rate	On-Time (%)	Median Closure Days
Fall	285482	92.89%	81.81%	0.38
Summer	319491	92.9%	81.44%	0.49
Spring	301888	92.96%	79.5%	0.49
Winter	292764	93.46%	75.02%	0.58

Organizational Patterns: Department Performance

Department-level analysis (Table 4) reveals substantial heterogeneity in service delivery performance. The Public Works Department (PWD) handled the largest case volume (641,037 cases) with 85.94% on-time performance and median closure time of 0.56 days. Boston Transportation Department (BTDT) achieved the highest on-time performance (87.76%) with exceptionally fast median closure (0.14 days). In contrast, the Inspectional Services Department (ISD) exhibited the poorest on-time performance (44.14%) with longest median closure time (2.87 days), indicating systematic operational challenges. Information and Communications (INFO) similarly struggled with 45.12% on-time performance. These patterns suggest department-specific factors substantially influence service delivery timelines.

Figure 3. On-Time Performance by Department

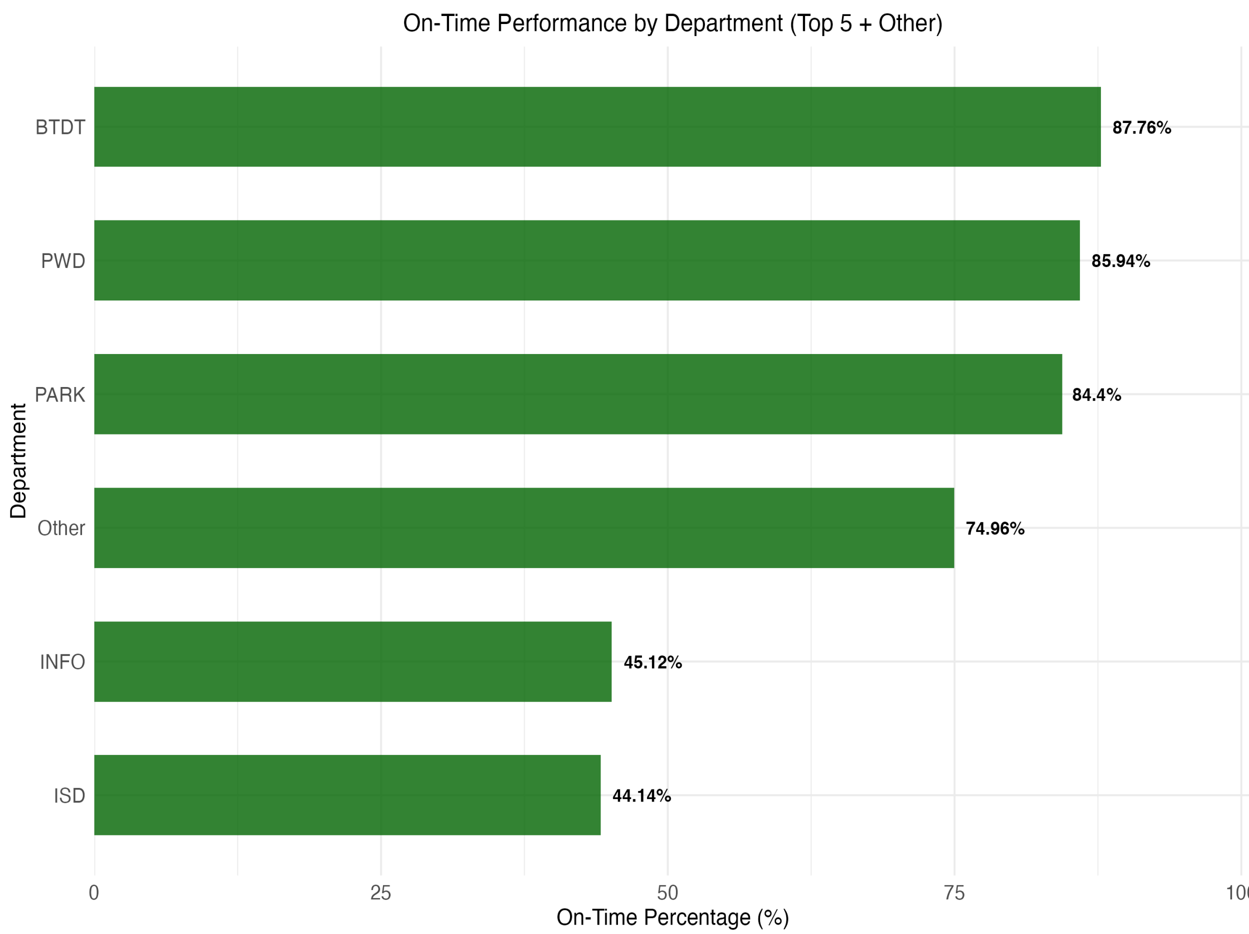


Table 4. Service Request Summary by Department

Department	Cases	On-Time (%)	Median Closure Days
PWD	641037	85.94%	0.56
BTDT	252013	87.76%	0.14
ISD	114889	44.14%	2.87
PARK	68907	84.4%	1.12
INFO	65608	45.12%	1.23
Other	57171	74.96%	0.15

Logistic Regression: Predicting Fast Closure

Binary logistic regression models were developed to classify service requests as fast (≤5 days) versus slow (>5 days) resolution. Table 5 presents results from four nested specifications. Model 1 (temporal factors only) achieved 86.7% test accuracy with log-likelihood of -248,937.8. Adding department indicators (Model 2) improved model fit (log-likelihood = -243,919.5) while maintaining accuracy. Incorporating request type (Model 3) substantially enhanced predictive power, reducing AIC to 386,097.5 despite marginally lower accuracy (86.5%). The full model (Model 4) including neighborhood and submission source achieved the best fit (AIC = 376,433.2) with 86.4% accuracy. The baseline odds ratio of 147.6 in the full model indicates strong baseline propensity for fast closure, with various factors modifying these odds Hosmer et al., 2013).

Table 5. Logistic Regression Models for Fast Closure Classification

	Dependent variable:			
	Fast Closure (<5 days = 1)			
	Temporal Only (1)	Add Department (2)	Add Request Type (3)	Full Model (4)
Constant	1.243 (0.01)	0.982 (0.021)	5.174 (0.12)	4.994 (0.161)
Odds Ratio (Constant)	**3.465**	**2.669**	**176.647**	**147.598**
Test Accuracy	**0.867**	**0.867**	**0.865**	**0.864**
Weekend	Yes	Yes	Yes	Yes

Season	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Department	No	Yes	Yes	Yes
Request Type	No	No	Yes	Yes
Neighborhood	No	No	No	Yes
Source	No	No	No	Yes
Observations	*647,835*	*647,835*	*647,835*	*647,835*
Log Likelihood	*-248,937.800*	*-243,919.500*	*-193,024.800*	*-188,182.600*
Akaike Inf. Crit.	*497,893.500*	*487,867.000*	*386,097.500*	*376,433.200*

Note: *'Yes' indicates variable category inclusion. AIC = Akaike Information Criterion (lower values indicate better fit).*

Model performance metrics (Table 6) demonstrate strong predictive capability. The full model achieved 86.37% accuracy, 87.59% precision, and exceptional 98.18% recall, yielding an F1 score of 0.926. The high recall indicates the model successfully identifies nearly all fast-closure cases, while precision of 87.59% suggests moderate false positive rates. The confusion matrix (Table 7) reveals the model correctly classified 236,216 fast cases and 3,596 slow cases, with 33,464 false positives (slow cases predicted as fast) and only 4,368 false negatives (fast cases predicted as slow).

Table 6. Logistic Regression Model Performance Metrics

Model	**Accuracy**	**Precision**	**Recall**	**F1_Score**
Model 4	86.37%	87.59%	98.18%	0.9259

Table 7. Confusion Matrix for Logistic Regression Classification

Classification	**Actual: Slow**	**Actual: Fast**
Predicted: Slow	3596	4368
Predicted: Fast	33464	236216

Linear Regression: Predicting Continuous Closure Time

Linear regression models were estimated to predict continuous closure time in days using the same nested specification approach. Table 8 presents results showing progressive improvement in explanatory power. Model 1 (temporal only) explained minimal variance ($R^2 = 0.022$) with constant of 2.691 days. Adding department indicators (Model 2) tripled explanatory power ($R^2 = 0.060$), while incorporating request type (Model 3) achieved substantial improvement ($R^2 = 0.238$). The full model including neighborhood and source variables explained 25.9% of variance with residual standard error of 2.537 days. All models exhibited highly significant F-statistics ($p < 0.001$), confirming systematic patterns beyond random variation. The progressive R^2 increase demonstrates that request-specific characteristics (type, department) contribute substantially more predictive value than temporal factors alone.

Table 8. Linear Regression Models for Closure Time Prediction

	Dependent variable:			
	Closure Time (Days)			
Constant	2.691 (0.009)	2.296 (0.011)	2.023 (0.02)	2.628 (0.025)
Weekend	Yes	Yes	Yes	Yes
Season	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Department	No	Yes	Yes	Yes
Request Type	No	No	Yes	Yes
Neighborhood	No	No	No	Yes
Source	No	No	No	Yes
Observations	*943,079*	*943,079*	*943,079*	*943,079*
R^2	*0.022*	*0.060*	*0.238*	*0.259*
Adjusted R^2	*0.022*	*0.060*	*0.238*	*0.259*
Residual Std. Error	*2.913 (df = 943070)*	*2.857 (df = 943065)*	*2.572 (df = 943055)*	*2.537 (df = 943045)*
F Statistic	*2,678.006*** (df = 8; 943070)*	*4,600.489*** (df = 13; 943065)*	*12,800.760*** (df = 23; 943055)*	*9,967.019*** (df = 33; 943045)*

***Note**: <0.1; **p<0.05; ***p<0.01; 'Yes' indicates the variable category is included in the model.*

LASSO Regression: Variable Selection and Comparison

LASSO regression was implemented to identify the most predictive features from nine candidate predictors. Table 9 compares OLS and LASSO results. The optimal penalty parameter ($\lambda = 0.024$) selected eight of nine predictors, eliminating only "Type Avg Closure" by shrinking its coefficient to zero. Both models achieved similar R^2 values (OLS: 0.1646, LASSO: 0.1639), indicating minimal loss in explanatory power despite reduced model complexity (Tibshirani, 1996).

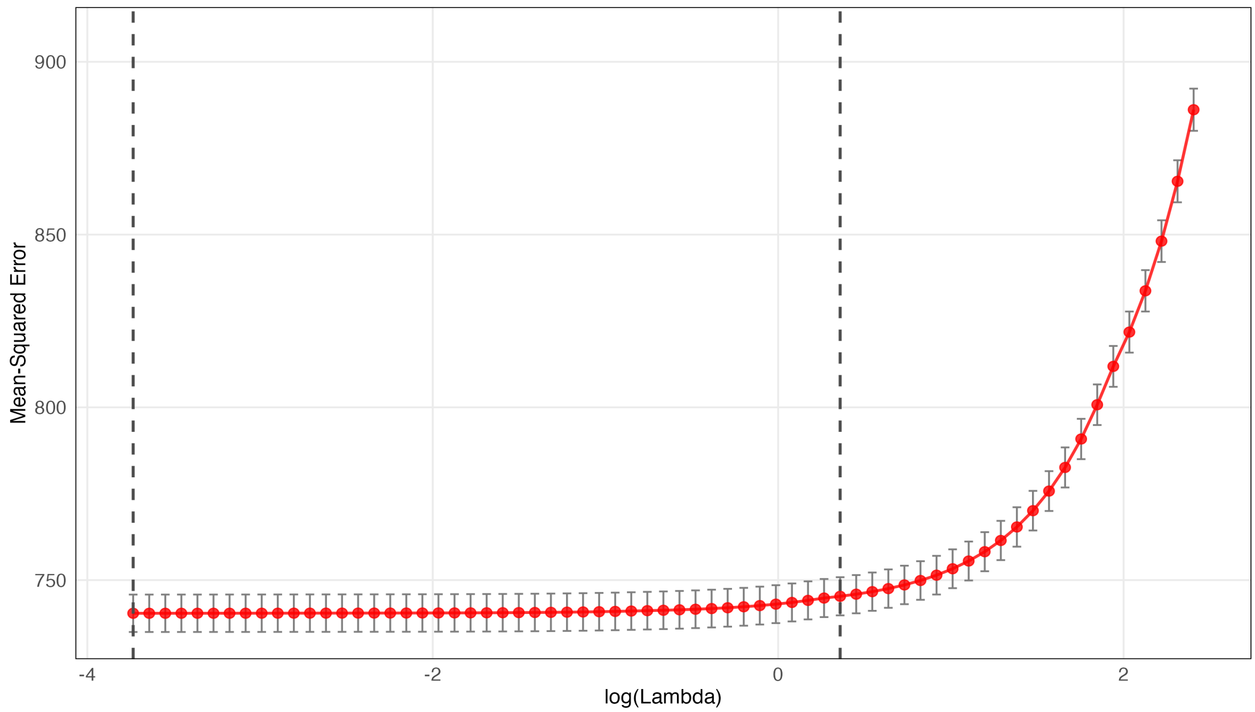
Key findings include Daily case volume exhibited negative association with closure time ($\beta = -0.0043$, $p < 0.001$), suggesting efficiency gains at higher volumes. Conversely, hourly case volume showed positive association ($\beta = 0.0042$, $p < 0.001$), indicating strain during peak periods. Department average closure time demonstrated the strongest effect ($\beta = 10.50$, $p < 0.001$), confirming department characteristics as primary determinants. Neighborhood average closure time also showed substantial positive association ($\beta = 1.17$, $p < 0.001$). Temporal variables (hour, day, week) contributed minimally, with LASSO shrinking most coefficients toward zero, consistent with earlier findings that temporal factors explain minimal variance.

Table 9. Comparison of OLS and LASSO Regression for SLA Duration

	Dependent variable:	
	SLA Duration (Days)	
	OLS Regression	LASSO Regression
Hour of Day	0.0038 (0.0082)	0.0000 (0.0082)
Day of Month	-0.0086** (0.0037)	-0.0059 (0.0037)
Day of Year	0.0046 (0.0165)	-0.0009 (0.0165)
Week of Year	-0.0402 (0.1156)	-0.00002 (0.1156)
Daily Case Volume	-0.0046*** (0.0001)	-0.0043*** (0.0001)
Hourly Case Volume	0.0065*** (0.0015)	0.0042*** (0.0015)
Neighborhood Avg Closure	1.2268*** (0.0956)	1.1677*** (0.0949)
Department Avg Closure	10.5195*** (0.0412)	10.5028*** (0.0382)
Type Avg Closure	2.6989*** (0.0193)	
Constant	-13.5847*** (0.2401)	-13.5673*** (0.2393)
Lambda	-	*0.023883*
Non-Zero Predictors	*9*	*8*
Observations	*685,220*	*685,220*
R^2	*0.1646*	*0.1639*
Residual Std. Error	*27.2099 (df = 685210)*	*27.2100 (df = 685211)*

***Note**: **p < 0.05; ***p < 0.001; indicates coefficient shrunk to zero by LASSO. Do not interpret LASSO coefficients for statistical significance because LASSO is primarily used for prediction and variable selection.*

Figure 4. LASSO Cross-Validation: Mean-Squared Error by Lambda



Interpretation and Conclusions

Key Findings

The resolution of Boston's 311 requests follows predictable temporal patterns. The median time to closure decreased significantly from ~1.00 day in 2015 to ~0.35 day in recent years. On-time closure had its highest value of 83.01% in 2016, but it has since decreased. Compared to the fall, on-time closure during the winter was consistently ~6% lower on average while median days to closure was consistently around 1.5 times slower. This could be caused by seasonal differences in workload or weather-related delays. Among other features, service requests were found to resolve fastest and most often on time when compared to the SLA deadline during summer.

There was also variation among different departments. The Inspectional Services Department (ISD) as well as Information & Communications (INFO) had the lowest rates of on-time closure (44–45%) and highest median days to closure while Public Works (PWD) and Boston Transportation Department (BTDT) had on-time closure rates of 85–88%. Adding department and request type as factors increased the R^2 value from .022 to .238. The accuracy of our logistic regression model (case will be fast ($\leq$ 5 days)) model was 86.4% with an F1 score of 0.926. When controlling for many variables, LASSO regression showed that department average close time was the most significant predictor of SLA length, highlighting the importance of department operations in setting closure time. These findings suggest that a request's creating department has more of an effect on time to closure than the time of year it was created.

Practical Implications

From these results we can draw some recommendations to enhance operations at Boston 311. Since ISD and INFO score significantly worse than other departments for all months, these departments should be focused on during workflow/staffing/process reviews to improve performance. These departments may use best practices from Public Works and Transportation to improve. Additionally, predictive modeling can aid in implementing a risk-based case management strategy. Cases that are predicted to close slowly can be identified at the beginning and monitored more closely or pushed to close quicker.

Because only a small amount of variation can be explained by the time of year, efforts to improve performance should not be focused on seasonal staffing increases or decreases. If higher case volume does correlate to faster case closure, there may be learning effects or economies of scale that can be utilized by leveling case workload. Lastly, because on-time performance has been decreasing since 2016 while average time to close has remained steady, SLAs should be reassessed. Departments may need to improve capacity or adjust SLAs to more realistic expectations.

Limitations and Future Research

Our study has a few shortcomings, each of which suggests an opportunity for future work. The relatively low R2 of the linear models suggests that key predictors of closure time are unobserved, and extending the model to account for additional information about each request—its complexity, the number of staff available to work on it, the weather, or features of the request text itself—may help. Additionally, this work does not differentiate between requests with larger public consequences should they be delayed; developing priority-weighted performance metrics is an area for further investigation.

The threshold for fast closure was determined by heuristic search; alternate thresholds or even time-to-event models may be more appropriate. Finally, we have only considered the accuracy of the models on past data; future work should determine their effectiveness when deployed, accounting for possible feedback.

Conclusion

In this project, we explore how combining various forms of analysis allows us to extract practical value out of a dataset containing millions of entries from city-wide 311 data in Boston from 2015-2019. Though aggregate on-time closure rates were high for most of the studied period, we found a downward trend beginning in 2016; though total time to close did not change much, there were proportionally fewer closures completed within the targeted response time. Response times also varied by season, with slower closure times in winter months. This suggests that while there may be scope for improving total closure times by modeling these seasonal trends, Boston 311 likely sees larger gains from adjusting workflows and staff schedules based on workload, rather than just time of day.

Further, we found that attributes at the department-level play a large role in how quickly requests are closed. Both our regression and regularization models showed that when controlling for request type and department, date-only attributes did little to explain the variance in total time to close. In fact, our model's R-squared increased by almost a factor of 10 when department and request type were included as features. The logistic regression model was also able to predict whether a given case would be closed quickly with high accuracy. From this initial exploration, we can conclude that there is strong potential to improve Boston 311's operations through identifying bottlenecks at the department-level, setting realistic SLAs, and predicting which cases will take a long time to close (Bluman, 2021).

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